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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CCT) (CBCS) Degree Examinations, December 2025

25CCT102 – CONSTRUCTION ECONOMICS & FINANCE

(For CCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) State and explain the Law of Supply and Law of Demand. Illustrate with suitable diagrams and explain the factors affecting each.	10	L2	1	1
	b) Zenith Associates has the following details: Fixed cost = Rs. 20,00,000 Variable cost per unit = Rs. 100 Selling price per unit = Rs. 200 Find (i) The break-even quantity, (ii) The break-even sales (iii) If the actual production quantity is 60,000, Find: (iv) contribution; and (ii) margin of safety by all methods	10	L3	1	1,3
2.	a) Explain the concept of Break-even Analysis. Derive the break-even point formula and discuss its significance in managerial decision-making.	10	L3	1	1
	b) In the design of a jet engine part, the designer has a choice of specifying either an aluminium alloy casting or a steel casting. Either material will provide equal service, but the aluminium casting will weigh 1.2 kg as compared with 1.35 kg for the steel casting. The aluminium can be cast for Rs. 80.00 per kg. and the steel one for Rs. 35.00 per kg. The cost of machining per unit is Rs. 150.00 for aluminium and Rs. 170.00 for steel. Every kilogram of excess weight is associated with a penalty of Rs. 1,300 due to increased fuel consumption. Which material should be specified and what is the economic advantage of the selection per unit?	10	L4	1	1,3
		Unit – II			
3.	a) Explain the steps involved in the planning of capital expenditures. How do engineers ensure that capital projects are technically and financially feasible?	10	L2	2	1
	b) Explain the steps involved in calculating the Internal Rate of Return (IRR) for a project with uniform annual cash inflows. The cost of a project is Rs. 32,400. It is expected to generate cash inflows of Rs. 16,000, Rs. 14,000 and Rs. 12,000 through its 3-year life period. Calculate the Internal Rate of Return of the Project.	10	L3	2	1,3
4.	a) Explain how risk and uncertainty affect capital budgeting decisions. What techniques are used to account for uncertainty in project evaluation?	10	L2	2	1
	b) List some of the main objectives of evaluation of public alternatives. In a particular locality of a state, the vehicle users take a roundabout route to reach certain places because of the presence of a river. This results in excessive travel time and increased fuel cost. So, the state government is planning to construct a bridge across the river. The estimated initial investment for constructing the bridge is Rs. 40,00,000. The estimated life of the bridge is 15 years. The annual operation and maintenance cost is Rs. 1,50,000. The value of fuel savings due to the construction of the bridge is Rs. 6,00,000 in the first year and it increases by Rs. 50,000 every year thereafter till the end of the life of the bridge. Check whether the project is justified based on BC ratio by assuming an interest rate of 12%, compounded annually.	10	L3	2	1,3

Unit – III

5. a) Explain the types of replacement problems encountered in engineering economics. Discuss the factors influencing replacement decisions. 10 L2 3 1
- b) A firm is considering replacement of an equipment, whose first cost is Rs. 4,000 and the scrap value is negligible at the end of any year. Based on experience, it was found that the maintenance cost is zero during the first year and it increases by Rs. 200 every year thereafter.
 (a) When should the equipment be replaced if $i = 0\%$?
 (b) When should the equipment be replaced if $i = 12\%$? 10 L3 3 1
6. a) Discuss the importance of replacement and maintenance analysis in engineering management. How do these analyses contribute to cost-effective operations? 10 L3 3 1
- b) Two years ago, a machine was purchased at a cost of Rs. 2,00,000 to be useful for eight years. Its salvage value at the end of its life is Rs. 25,000. The annual maintenance cost is Rs. 25,000. The market value of the present machine is Rs. 1,20,000. Now, a new machine to cater to the need of the present machine is available at Rs. 1,50,000 to be useful for six years. Its annual maintenance cost is Rs. 14,000. The salvage value of the new machine is Rs. 20,000. Using an interest rate of 12%, find whether it is worth replacing the present machine with the new machine. 10 L4 3 1,3

Unit – IV

7. a) What is an Opening Entry? Explain the procedure for recording opening entries and discuss its significance in accounting 10 L2 4 1
- b) What are the various objectives and functions of accounting? Classify the following under Personal, Real and Nominal accounts:
 Stock, Loan, Insurance, Salary, Interest, Bank, Cash, Capital, Prepaid Interest, Salary outstanding, Drawing, Bank overdraft, Salary prepaid
 Fixtures, Bills receivable, Machinery, Building, Goodwill. 10 L3 4 1,2
8. a) Explain the different methods of preparing a trial balance. Discuss the possible errors that do not affect the agreement of a trial balance. 10 L3 4 1
- b) Define a Balance Sheet. Discuss the classification of assets and liabilities and explain the format of a balance sheet with a suitable example. 10 L3 4 1

Unit – V

9. a) Explain the preparation of Profit and Loss Account. Discuss how it helps in determining the operational efficiency of a business. 10 L3 5 1
- b) What are cash flows and fund flows? Explain the differences between cash flow statement and fund flow statement with examples. 10 L2 5 1
10. a) How does the adoption of IFRS improve the comparability and transparency of financial statements globally? Discuss with examples. 10 L3 5 1
- b) Define Working Capital Management. Discuss its importance in ensuring the liquidity and profitability of an organization. 10 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
